Placebo Trial Design Classroom Run Kit

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	**Time**	30-40 min
Difficulty	Medium	**ID**	placebo-trial-design

Big idea: Controls protect knowledge from bias.

One-Minute Teacher Brief

Students design a method that protects knowledge from expectation and bias.

Student Prompt

Inspect Placebo Trial Design Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Placebo Trial Design Stimulus A	Student-facing stimulus 1: A mock treatment trial with control, placebo, randomisation, and blinding cards. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Placebo Trial Design Stimulus B	Student-facing stimulus 2: A flawed trial students must repair before trusting its conclusion. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Placebo Trial Design Reveal / Twist	Student-facing stimulus 3: A debrief on ethics when belief itself changes outcomes. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Controls protect knowledge from bias.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Placebo Trial Design Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Placebo Trial Design Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Why is personal experience not enough in science?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how controls protect knowledge from bias.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

• Work silently for the first response so you can notice your own starting point.

• Record one knowledge claim, the evidence behind it, and your confidence level.

• After the reveal or comparison, update your claim in a different colour or column.

• Use the debrief to answer: Why is personal experience not enough in science?

Setup Checklist

• Prepare a starter stimulus using: A mock treatment trial with control, placebo, randomisation, and blinding cards.

• Print or project the student response grid before the activity begins.

• Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

• A mock treatment trial with control, placebo, randomisation, and blinding cards.

• A flawed trial students must repair before trusting its conclusion.

• A debrief on ethics when belief itself changes outcomes.

Resources To Use

• Resource pack: resources/placebo-trial-design/index.html

• Card deck CSV: resources/placebo-trial-design/cards.csv

• Visual prompt: resources/placebo-trial-design/visual-prompt.svg

• Printable student worksheet with observation, inference, evidence, and confidence columns.

• Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.

• Classroom run kit with prompt cards, example stimuli, and debrief moves.

• PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Groups design a fair test of the product.

2. They must include a control group, placebo, blinding, outcome measure, sample size and ethics note.

3. Groups exchange designs and find weaknesses.

4. The class agrees minimum standards for a trustworthy test.

Debrief Moves

• Knowledge question: Why is personal experience not enough in science?

- How do controls improve knowledge?
- Can scientific methods remove all bias?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...